

HW3

Greyson Knippel
CPSC-321- Fall 2026

October 5, 2026

Query 1

thresholds: province area > 100000, country gdp > 40000.

output

country_code	country_name	gdp	province_name	area
US	United States of America	46900	Oregon	254799
US	United States of America	46900	Washington	184661
CA	Canada	43100	Ontario	1076395
CA	Canada	43100	British Columbia	944735

(4 rows)

matching example

row: US, Oregon.

country: ('US', 'United States of America', 46900, 3.8)

province: ('Oregon', 'US', 254799)

oregon's area is $254799 > 100000$ and the US gdp is $46900 > 40000$.

nonmatching example

country: ('MX', 'Mexico', 15300, 4.7)

province: ('Sonora', 'MX', 179355)

sonora's area is over the threshold, but mexico's gdp is $15300 < 40000$. sonora is not in the output.

Query 2

thresholds: same as query 1.

output

country_code	country_name	gdp	province_name	area
US	United States of America	46900	Oregon	254799
US	United States of America	46900	Washington	184661
CA	Canada	43100	Ontario	1076395
CA	Canada	43100	British Columbia	944735

(4 rows)

examples

same as query 1. the output is identical to query 1.

Query 3

thresholds: province area < 100000, city population between 10000 and 300000 (inclusive).

output

country_code	province_name	area
DE	Bavaria	70550
DE	Saxony	18450

(2 rows)

matching example

row: DE, Bavaria.

province: ('Bavaria', 'DE', 70550)
city: ('Neustadt', 'Bavaria', 'DE', 13000)

bavaria's area is 70550 < 100000 and neustadt's population of 13000 is in the range.

nonmatching example

province: ('Jalisco', 'MX', 78599)
city: ('Guadalajara', 'Jalisco', 'MX', 1385629)
city: ('Zapopan', 'Jalisco', 'MX', 1476491)
city: ('Tlaquepaque', 'Jalisco', 'MX', 687127)

jalisco's area is under the threshold, but every city is over 300000. jalisco is not in the output.

Query 4

thresholds: same as query 3.

output

country_code	province_name	area
DE	Bavaria	70550
DE	Saxony	18450

(2 rows)

examples

same as query 3. the output is the same as query 3.

Query 5

thresholds: city population > 500000, country inflation < 4.0.

output

country_code	country_name	inflation	province_name	area
CA	Canada	2.4	Ontario	1076395
DE	Germany	2.2	Bavaria	70550
DE	Germany	2.2	Saxony	18450

(3 rows)

matching example

row: CA, Ontario.

```
country: ('CA', 'Canada', 43100, 2.4)
province: ('Ontario', 'CA', 1076395)
city:     ('Toronto', 'Ontario', 'CA', 2794356)
city:     ('Ottawa', 'Ontario', 'CA', 1017449)
```

ontario has two different cities over 500000 and canada's inflation is $2.4 < 4.0$.

nonmatching example

```
country: ('US', 'United States of America', 46900, 3.8)
province: ('Washington', 'US', 184661)
city:     ('Seattle', 'Washington', 'US', 737015)
city:     ('Spokane', 'Washington', 'US', 228989)
city:     ('Vancouver', 'Washington', 'US', 190915)
```

the US inflation is under the threshold, but only seattle is over 500000. washington is not in the output.

Query 6

thresholds: same as query 5.

output

country_code	country_name	inflation	province_name	area
CA	Canada	2.4	Ontario	1076395
DE	Germany	2.2	Bavaria	70550
DE	Germany	2.2	Saxony	18450

(3 rows)

examples

same as query 5. the output is the same as query 5.

Query 7

no thresholds.

output

city_name_1	province_name_1	country_code_1	population_1	city_name_2	province_name_2	country_code_2	population_2
Vancouver	Washington	US	190915	Vancouver	British Columbia	CA	662248
Portland	Ontario	CA	15000	Portland	Oregon	US	652503

(2 rows)

matching example

row: Vancouver (US), Vancouver (CA).

city: ('Vancouver', 'Washington', 'US', 190915)

city: ('Vancouver', 'British Columbia', 'CA', 662248)

same name, different countries, different populations. the less populous city (US) is first.

nonmatching example

city: ('Neustadt', 'Bavaria', 'DE', 13000)

city: ('Neustadt', 'Saxony', 'DE', 12000)

same name, but both are in DE. neustadt is not in the output.

Query 8

threshold: border length > 400.

output

country_code	country_name
CA	Canada
US	United States of America

(2 rows)

matching example

row: US.

country: ('US', 'United States of America', 46900, 3.8)

country: ('MX', 'Mexico', 15300, 4.7)

border: ('MX', 'US', 3145)

the US has a higher gdp ($46900 > 15300$) and lower inflation ($3.8 < 4.7$) than mexico, and the border is $3145 > 400$. the US is in `country_code_2`.

nonmatching example

```
country: ('MX', 'Mexico', 15300, 4.7)
border:  ('MX', 'US', 3145)
border:  ('CA', 'MX', 500)
```

mexico has a lower gdp and higher inflation than both of its neighbors. mexico is not in the output.

Query 9

threshold: same as query 8.

output

country_code	country_name
US	United States of America
CA	Canada

(2 rows)

examples

same as query 8. the output has the same rows as query 8 (order differs, no sorting required).

Query 10

question: which provinces have a city with over 1000000 people, in a country with inflation under 4.0?

thresholds: city population > 1000000, country inflation < 4.0.

sorted by province area, largest first, so the largest regions are listed first.

output

country_name	province_name	area
Canada	Ontario	1076395
Germany	Bavaria	70550

(2 rows)

matching example

row: Germany, Bavaria.

```
country: ('DE', 'Germany', 52100, 2.2)
province: ('Bavaria', 'DE', 70550)
city:     ('Munich', 'Bavaria', 'DE', 1512491)
```

munich is over 1000000 and germany's inflation is $2.2 < 4.0$.

nonmatching example

```
country: ('MX', 'Mexico', 15300, 4.7)
province: ('Jalisco', 'MX', 78599)
city:     ('Guadalajara', 'Jalisco', 'MX', 1385629)
city:     ('Zapopan', 'Jalisco', 'MX', 1476491)
```

jalisco has cities over 1000000, but mexico's inflation is $4.7 > 4.0$. jalisco is not in the output.